

Principles of Inheritance and Variation

TOPIC - Inheritance of Two Genes, Polygenic Inheritance , Chromosomal Theory

1. Select odd one trait w.r.t. non-allelic genetic interactions.

- (1) Flower colour in sweet pea
- (2) Fruit colour in summer squash
- (3) Shepherd's purse capsule shape
- (4) Seed shape in pea

2. Given below are two statements:

Assertion(A) : The law of independent assortment can be studied by means of dihybrid cross.

Reason(R) : The law of independent assortment is applicable only to linked genes.

- (1) Both Assertion (A) and Reason (R) are correct and Reason (R) is a correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are correct but Reason (R) is not a correct explanation of Assertion (A).
- (3) Assertion (A) is correct and Reason (R) is incorrect.
- (4) Assertion (A) and Reason (R) both are incorrect.

3. The numbers of phenotypes and genotypes in F_2 generation of a Mendelian dihybrid cross are

- (1) phenotypes 4: genotypes 16
- (2) phenotypes 4: genotypes 8
- (3) phenotypes 9: genotypes 4
- (4) phenotypes 4: genotypes 9

4. Given below are two statements:

Statement-I : The genotype with three dominant alleles and three recessive alleles will have a dark skin colour.

Statement II : Phenylketonuria, expression characterised by mental retardation and a reduction in hair and skin pigmentation.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both statements I and II are correct.
- (2) Statement I is correct but statement II is incorrect.
- (3) Statement I is incorrect but statement II is correct.
- (4) Both statements I and II are incorrect.

5. Which of the following genotypes of *Lathyrus odoratus* is/are not able to produce purple colour in flowers?

- (1) ccPp
- (2) CCpp
- (3) CcPp
- (4) Both (1) & (2)

6. What is the skin colour of a human being with the genotype AAbbCc?

- (1) Fairly dark
- (2) Mulatto
- (3) Darly
- (4) Very light

7. Mendel's work remained unrecognized for many years. Find out the true reason for the same.

(I) Mendel's concept of genes was not accepted by his contemporaries as an explanation for the continuous variation seen in nature.

(II) The approach of using mathematics was new and unacceptable by other biologists.

(III) He could not provide any physical proof for the existence of factors.

(IV) Communication was not easy in those days and his work could not be widely published.

Choose the most appropriate answer

- (1) I and II
- (2) II and III
- (3) III and IV
- (4) All of these

8. Who proposed chromosomal theory of inheritance independently?

- (1) Sutton and Boveri
- (2) Meselson and Stahl
- (3) Watson and Crick
- (4) T.H. Morgan

9. Genes which are present on the same chromosome

- (1) do not form any linkage group.
- (2) affect the phenotype by forming interactive groups.
- (3) form a linkage group.
- (4) form different groups depending upon their relative distance.

10. Phenylketonuria is an example of:

- (i) Pleiotropy**
- (ii) Genetic disorder**
- (iii) Metabolic disorder**

- (1) (i) and (ii)
- (2) (i) and (iii)
- (3) All three
- (4) Only (ii) and (iii)

11. Round & yellow seeded pea plant (RRWW) was crossed with wrinkled & green seeded pea plant (rrww). F_1 generation is formed. On selfing of F_1 generation, F_2 generation is obtained. How many plants will be green seeded pea plants in F_2 ?

- (1) 1 / 16
- (2) 3 / 16
- (3) 4 / 16
- (4) 1 / 16

12. Given below are two statements:

Assertion (A) : The inheritance of human blood group represents multiple allelism.

Reason (R) : In polygenic inheritance, a trait is controlled by three or more than three genes.

- (1) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are true but Reason (R) is not a correct explanation of Assertion (A).
- (3) Assertion (A) is true and Reason (R) is false.
- (4) Assertion and Reason (R) both are false.

13. Read the following statements and choose correct option

(A) A gamete carries only one factor of a character

(B) Starch synthesis in wrinkled seeded pea plants is most efficient

(C) Modified allele is always the recessive allele

(1) All are incorrect except A

(2) All are correct except B

(3) All are incorrect

(4) All are correct

14. Dihybrid phenotypic ratio in F₂ generation of collaborative gene interaction is

(1) 9 : 3 : 4

(2) 9 : 3 : 3 : 1

(3) 9 : 7

(4) 9 : 6 : 1

15. Given below are the two statements:

Assertion: Morgan coined the term linkage to describe the physical association of genes on a chromosome.

Reason: Linkage shows more non-parental type combination of genes.

Choose the most appropriate answer

(1) Both assertion and reason are true and the reason is the correct explanation of assertion.

(2) Both assertion and reason are true but the reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Both assertion and reason are false.

16. What is polygenic inheritance?

(1) The inheritance of traits controlled by a single gene.

(2) The inheritance of traits controlled by three or more genes.

(3) The inheritance of traits controlled only by environmental factors.

(4) The inheritance of traits with no genetic basis.

17. What will be the number of genotypes in F₂ generation if a character is controlled by 4 pairs of polygenes?

(1) 16

(2) 9

(3) 25

(4) 81

18. Given below are the two statements:

Assertion: Mendel proposed the law of independent assortment on the basis of results of dihybrid cross.

Reason: When two pairs of traits are combined in a hybrid, segregation of one pair of characters is independent of the other pair of characters.

Choose the most appropriate answer from the options given below:

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true but reason is not correct explanation of assertion

(3) Assertion is true, but reason is false.

(4) Both the assertion and the reason are false.



19. In a Dihybrid cross, if you get 9:3:3:1 ratio it denotes that:

- (1) the alleles of 2 genes are interacting with each other.
- (2) it is multigenic inheritance.
- (3) it is a case of multiple allelism.
- (4) the alleles of 2 genes are segregating independently..

20. Fill in the blanks and choose correct option.

I. In _[A]___ inheritance, the dominant alleles of the genes show cumulative effect.

II. First linkage map was developed in _[B]_____.

III. Chromosome __[C]_____ is called Philadelphia chromosome.

- (1) A – Polygenic, B – Neurospora, C – 9
- (2) A – Qualitative, B – Pisum, C – 5
- (3) A – Quantitative, B – Drosophila, C – 22
- (4) A – Monogenic, B – Lathyrus, C – 21

ANSWER KEY

1. (4) 2. (3) 3. (4) 4. (3) 5. (4) 6. (2) 7. (4) 8. (1) 9. (3) 10. (3)
11. (3) 12. (2) 13. (1) 14. (2) 15. (3) 16. (2) 17. (4) 18. (2) 19. (4) 20. (2)

SEEP PAHUJA
BIOLOGY
KEEP WALKING

